

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 7 Summer 2025 (Supply.)

Course :Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTEEOE705B - ELECTRIC AND HYBRID ELECTRIC VEHICLES

Time : 2 hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

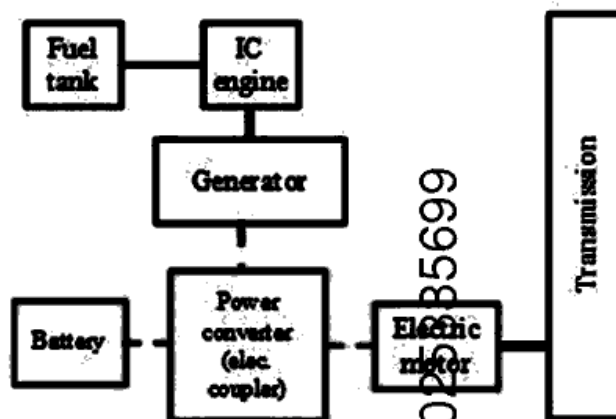
1 A hybrid vehicle combines any type of two power (energy) sources. Possible combinations include

- A) Diesel/ electric. B) gasoline/ fly wheel.
C) fuel cell (FC)/battery. D) All Above.

When the vehicle moves, it encounters a resistive force that tries to retard its motion. The resistive forces are.

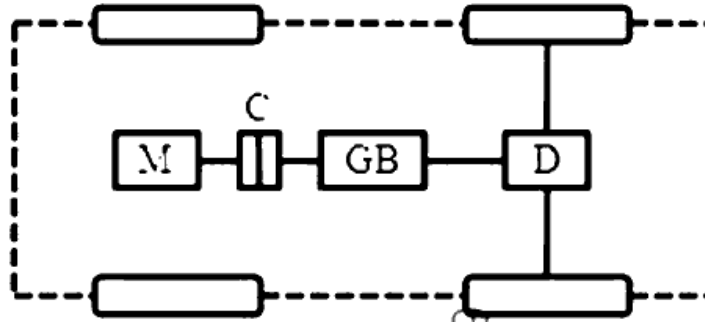
- A) Rolling resistance. B) Aerodynamic drag. C) Uphill resistance. D) All Above.

3 The figure shown below indicates which of the following HEV configuration?



- A) Series Hybrid B) Parallel Hybrid C) Combined Hybrid D) All Above

4 Identify type of Alternative configuration of EV based on



A) Drive Train. B) Power Source. C) Both A & B. D) None of Above

5 The fundamental requirement for traction _____ used in EVs is to generate propulsion torque over a wide speed range.

A) Gear Box. B) Motor C) Clutch D) All above

6 In a DC motor drive, which of the following techniques is most commonly used to control the speed of the motor?

A) Adjusting the frequency of the input supply. B) Changing the flux linkage.
C) Varying the voltage supplied to the armature. D) Using phase-angle control

During regenerative braking, which component primarily converts kinetic energy into electrical energy?

A) Battery. B) Electric motor acting as a generator.
C) Power controller. D) Energy recovery circuit.

8 What is the primary byproduct of a hydrogen fuel cell during energy generation?

A) Carbon dioxide. B) Oxygen. C) Water. D) Nitrogen.

9 What is the main component that stores electrical energy in an electric vehicle?

A) Radiator. B) Battery pack. C) Carburetor. D) Exhaust pipe

10 Which type of battery is commonly used in modern electric vehicles due to its high energy density and efficiency?

A) Nickel Cadmium (NiCd). B) Lead Acid. C) Lithium-Ion (Li-ion). D) Alkaline

In a pure electric vehicle, the primary focus of energy management strategies is on

A) Maximizing battery cycle life and range.
B) Improving thermal stability of the motor.
C) Reducing the weight of the battery pack.
D) Synchronizing multiple motors for higher efficiency.

12 In rule-based energy management strategies, decisions are typically based on:

A) Predefined conditions and simple logic rules.
B) Complex optimization algorithms.
C) Machine learning predictions.
D) Real-time fuel consumption monitoring

Q2. Solve the following.

A) Economic and Environmental Impact of Electric Hybrid Vehicle

6

B) Compare the performance of EV and HEV

6

- Q3. Solve the following.
- A) Explain different power flow control in series HEVs with neat sketch 6
 - B) Draw electric vehicle drive train configurations based on power source and brief it 6
- Q4. Solve Any Two of the following.
- A) Describe the principal of operation induction motor with Mathematical treatment 6
 - B) What is Switch Reluctance motor drives with advantages 6
 - C) Explain the forward motoring and regenerative (forward)Braking control of a dc motor with single chopped, Give Circuit diagram and Show quadrants of Operation 6
- Q5. Solve Any Two of the following.
- A) Describe the phenomenon of state of Charge and Depth of Charge 6
 - B) What are the different novel alternative energy sources may be used in electric vehicle 6
 - C) Mathematical Modeling of Lead acid Battery 6
- Q6. Solve Any Two of the following.
- A) Describe the function of Control system in HEVs and EVs and elements of control theory 6
 - B) Show the core components of the fuzzy logic controller and the input and output relevant to HEVs 6
 - C) Explain typical control architecture of HEV for The Electronic Control Unit(ECU) 6

*** End ***